1. Challenge 1: Cloud Gaming Chaos

   A cloud gaming provider, like Novidia G-Force Now, hosts dedicated streaming servers for an esports tournament at the beginning of each month. Assume that the number of connection drops for a server in a month follows the Poisson distribution with mean 1.9. Suppose there are totally 15 servers allocated for the tournament, and the performance of a server in a month is regarded as unacceptable if there are more than 2 connection drops in that month. Assume that the monthly numbers of connection drops of the servers are independent.

   (a) (2 points) Find the probability that the performance of a randomly selected server in a certain month is unacceptable.

   (b) (2 points) For a certain server, find the probability that June of 2026 is the 3rd month in 2026 such that the performance of that server is unacceptable.

   (c) (2 points) Find the expected total number of unacceptable server performances for all 15 servers over one year.

   (d) (4 points) In order to assure the stream quality, the tournament organizers introduce the following term in the new hosting contract for the 15 servers, which will be effective on 1st January 2027.

   For each server, if its performance is unacceptable for 3 consecutive months in the new contract period, one hardware replacement order will be immediately issued to the provider, provided that no hardware replacement order has been issued for that server before.

   Find the probability that 3 or more hardware replacement orders will be issued to the provider on or before 30th April 2027.

2. Challenge 2: The VR Lag Spikes

The number of lag spikes in a day while playing on a newly released Beta Quest 3 VR headset follows the Poisson distribution with mean 4.8. Assume that the daily numbers of lag spikes are independent.

(a) (2 points) Find the probability that there are not more than 3 lag spikes in a day.

(b) (2 points) Find the probability that, in 3 consecutive days, there are at most 2 days with not more than 3 lag spikes in each day.

(c) A day is called a rage day if there are more than 5 lag spikes in that day; otherwise it is called a smooth day.

i. (2 points) Suppose today is a rage day. Find the mean number of smooth days between today and the next rage day.

ii. (2 points) Find the probability that the last day of a week is the third rage day in that week.

iii. (2 points) Find the probability that there are at least 4 consecutive rage days in a week.

3. Challenge 3: Gacha Grinding

The number of Wishes (digital loot boxes) opened by a player in the popular gacha game Jenshin Impact in a minute can be modelled by a Poisson distribution with a mean of 3.2. The probability distribution of the number of "Masterless Star-glitters" dropped per Wish is shown in the following table:

Number of Star-glitters dropped	1	2	3	4	5	6	≥ 7
Probability	0.12	0.7	0.08	0.04	0.03	0.02	0.01

(a) (3 points) Find the probability that fewer than 4 Wishes are opened by a player in a certain minute.

(b) (2 points) Find the probability that the 8th Wish opened is the 3rd Wish that drops exactly 2 Masterless Star-glitters.

(c) (2 points) Find the probability that exactly 3 Wishes are opened in a certain minute and each of them drops exactly 2 Masterless Star-glitters.

(d) (3 points) Find the probability that exactly 3 Wishes are opened in a certain minute and they drop a total of 6 Masterless Star-glitters.

4. Challenge 4: The Live Event Login Queue

Tom joins the login queue for a massive live event in Fort-night (like the Chapter-ending Big Bang event) at 7:10 PM. The Epoch Games server lets players in via login waves. A wave occurs at 7:20 PM and another wave occurs at 7:30 PM. The probability that Tom can successfully bypass the queue in a wave is 0.9 each time. If Tom gets in at 7:20 PM, the probability for him to miss the opening cinematic (be late to the event) is 0.1. If Tom gets in at 7:30 PM, the probability for him to be late to the event is 0.4. Tom will definitely be late if he cannot get through the queue in these two waves.

(a) (2 points) Find the probability that Tom successfully bypasses the queue on or before 7:30 PM on a certain day.

(b) (2 points) Find the probability that Tom is late to the event on a certain day.

(c) (2 points) Find the probability that Tom is late to the event 2 times out of 6 event days.

(d) There are 7 players in a Dischord voice channel, including Tom, waiting for a match. If Tom is late to the event, he gets tilted and will play Casual Mode; otherwise he will play Ranked League. The probabilities for each of the other 6 players to play Casual Mode and Ranked League are 0.7 and 0.3 respectively. Once the event ends, the 7 players lock in their game modes. No player changes their mode afterwards.

   i. (2 points) Find the probability that the 7 players are playing the same game mode.

   ii. (2 points) Find the probability that exactly 3 players are playing Ranked League.